Department of Mathematics, BIT Mesra, Ranchi

MA 107: Mathematics-II - UG

Instructions: Attempt all questions. Time: one hour

1. If the partial differential equation $(x-1)^2u_{xx}-(y-2)^2u_{yy}+2xu_x+2yu_y+2xyu=0$ is parabolic in $S \subseteq R^2$ but not in R^2/S, then S is
   (a) $\{ (x,y) \in R^2 : x=1 \, and \, y=2\}$ (b) $\{ (x,y) \in R^2 : x=1 \, or \, y=2\}$
   (c) $\{ (x,y) \in R^2 : x=1\}$ (d) $\{ (x,y) \in R^2 : y=2\}$

2. The general integral/solution of the partial differential equation

$$(y+zx)z_x-(x+yz)z_y=x^2-y^2$$

   is
   (a) $\{ F(x^2+y^2+z^2, xy-z)=0\}$ (b) $\{ F(x^2-y^2-z^2, xy+z)=0\}$
   (c) $\{ F(x^2+y^2+z^2, xy+z)=0\}$ (d) $\{ F(x^2+y^2-z^2, xy+z)=0\}$

3. Consider the following statements:
   P: If $u=f(x+iy)+g(x-iy)$, where the functions f and g are arbitrary, then $u_{xx}+u_{yy}=0$.
   Q: If $z=f(x^2-y)+g(x^2+y)$, where the functions f and g are arbitrary, then $z_{xx}-\frac{1}{x}z_x=4x^2z_{yy}$.
   (a) P is true, Q is false (b) P is false, Q is true
   (c) Both P and Q are true (d) Both P and Q are false

4. Let $u(x,t)$ be the solution of the wave equation $u_{tt}=u_{xx}$, $0<x<\pi$, $t>0$ with the initial conditions

$$u(x,0)=\sin x+\sin 2x+\sin 3x, \quad u_t(x,0)=0, \quad 0<x<\pi$$

   and boundary conditions $u(0,t)=u(\pi,t)=0$ for $t>0$. Then the value of $u(\frac{\pi}{2},\pi)$ is
   (a) 0 (b) $-\frac{1}{2}$ (c) 1 (d) -1

5. If $f(x)=x$, $0<x<\dfrac{\pi}{2}$ and $f(x)=\pi-x$, $\dfrac{\pi}{2}\le x<\pi$. Then choose the Fourier sine series of f in the given interval from the options.
   (a) $\dfrac{2}{\pi}\sum_{n=1}^{\infty}\dfrac{1}{n}\sin(n\pi)\sin(\dfrac{n\pi}{2})$. (b) $\sum_{n=1}^{\infty}\dfrac{4}{\pi n^2}\sin(\dfrac{n\pi}{2})\sin(nx)$.
   (c) $\sum_{n=1}^{\infty}\dfrac{2}{\pi^2n^2}\sin(\dfrac{n\pi}{2})\sin(\dfrac{nx}{2})$. (d) $\sum_{n=1}^{\infty}\dfrac{4}{\pi^2n^2}\sin(\dfrac{n\pi}{2})\sin(nx)$.

6. The diffusion equation

$$\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial^2 x}$$

   $u(0,t)=u(\pi,t)=0$ and $u(x,0)=\cos x\sin 5x$, admits the solution
   (a) $\frac{e^{-36t}}{2}(\sin 3x+e^{15t}\sin 5x)$ (b) $\frac{e^{-36t}}{2}(\sin 5x+e^{35t}\sin x)$
   (c) $\frac{e^{-36t}}{2}(\sin 6x+e^{20t}\sin 4x)$ (d) $\frac{e^{-36t}}{2}(\sin 4x+e^{20t}(\sin 4x+\sin 6x))$

7. Consider the following statements:
P: Fourier series of $f(t) = t, -\pi < t \leq \pi$ is

$$2(\sin t - \frac{1}{2}\sin 2t + \frac{1}{3}\sin 3t - \dots)$$

Q: Fourier series of $f(t) = t, 0 < t < 2\pi$ is

$$\pi - 2(\sin t + \frac{1}{2}\sin 2t + \frac{1}{3}\sin 3t + \dots)$$

(a) P is true, Q is false
(b) P is false, Q is true
(c) Both P and Q are true
(d) Both P and Q are false

8. Consider the heat equation

$$u_t = u_{xx}, \qquad 0 < x < \pi, \quad t > 0$$

with the boundary conditions $u(0,t) = 0$, $u(\pi,t) = 0$ for $t > 0$, and the initial condition $u(x,0) = \sin x$. Then $u(\frac{\pi}{2}, 1)$ is

(a) $\frac{e}{2}$
(b) $\frac{2}{e}$
(c) $\frac{1}{e}$
(d) e

9. PDE generated from the given relation $2z = (ax+y)^2 + b$, where x, y are independent variables and z is a dependent variable. If $p = z_x$, $q = z_y$ then

(a) $q^2 = py$.
(b) $p^2 = qx + py$.
(c) $q^2 = px + qy$.
(d) none of (a),(b) and (c) is correct

10. Consider the periodic function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \sin\left(x + \frac{\pi}{64}\right)$$

In the Fourier series expansion of f, with respect to the period $2L$, the Fourier coefficients are c (Constant term), $a_n, b_n, \forall n \geq 1$, the value of $\left[\left(\frac{c}{4}\right)^2 + \frac{1}{64}(a_1^2 + 64a_2^2 + b_1^2 + 32b_2^2)\right.$

(a) $\frac{1}{32}$
(b) 0
(c) 1
(d) none of the above